CS & DA

2024

Database Management System

ER Model and Conversion from ER Model to Relational Model

DPP-01 Discussion Notes



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[MCQ]

SQL —

#Q. ER model is

A. Physical design — disk

B. Logical design — Implement using codes

C. Conceptual design

D. None of the above

[NAT]

#Q. Consider the following tables T1 and T2.

In table T1 P is the primary key and Q is the foreign key referencing R in table T2 with on-delete cascade and on-update cascade. In table T2, R is the primary key and S is the foreign key referencing P in table T1 with on-delete set NULL and on-update cascade. In order to delete record (3,8) from the table T1, the number of additional records that need to be deleted from table T1 is ____.

0

T1

P	Q
2	2
3	8
7	3
5	8
6	9
8	5
9	8

T2

R	S
2	2
8	3
3	2
9	7
5	7
7	2

NULL

[NAT]

#Q.Consider a relational table R(A, B) as given below. A is the primary key of relation R and B is the foreign key referring to primary key A of relation R with on delete cascade. If we delete tuple (2, 3) from relation R, then total number of tuples (including (2, 3)) deleted from R to preserve referential integrity is_____

A	B
5	8
3	2
8	7
1	4
2	3
6	3
7	9
9	5
4	3



[MSQ]

#Q. Which of the following is/are true for an ER model?

partial

1 row

A. Weak entity must have total participation in identifying relation.

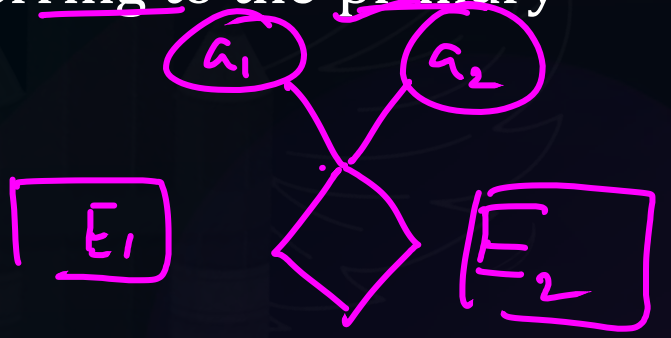
B. Entity corresponding to 1 side will include foreign key referring to the primary key of many side entity.

foreign

C. Descriptive attributes are associated with entity.

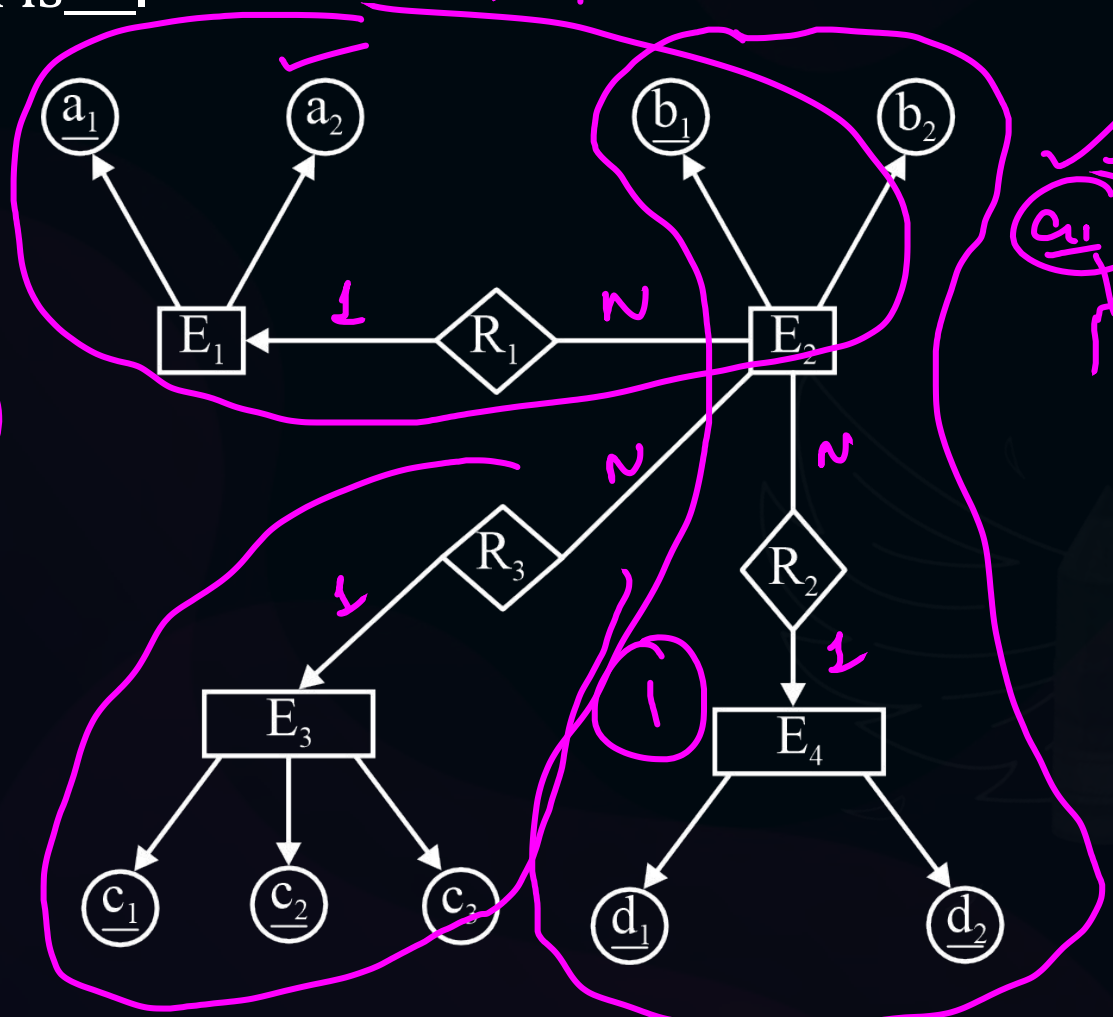
Relationship

D. Minimum cardinality of '1' specifies total participation.



[NAT]

#Q. Minimum number of tables required to convert the ER diagram into relational model is___.

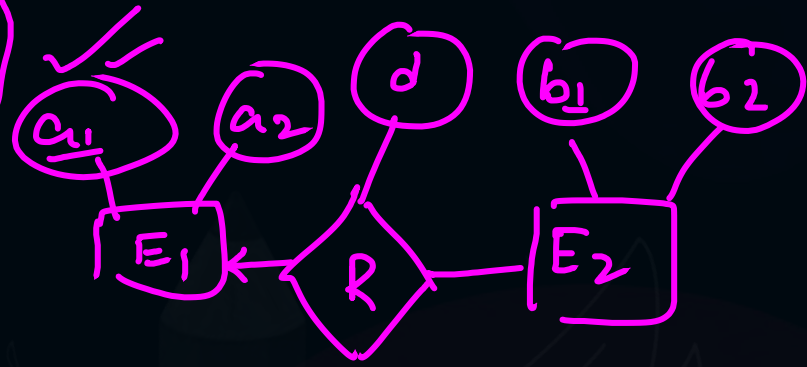


1+1 = 2

$E_4(\underline{d_1}\ \underline{d_2})$ X

$E_2(\underline{b_1}\ b_2, \underline{d_1}\ \underline{d_2})$

2+1 = 3



2 tables

$E_1(\underline{a_1}\ a_2)$

$E_2(\underline{b_1}, \underline{b_2}, a_1)$

THANK - YOU

